SWINBURNE UNIVERSITY OF TECHNOLOGY
SARAWAK CAMPUS

COS10005
Web Development
Lecture 6

Responsive CSS

KNOWING

Contents

- Responsive Web Design
- Approaches to responsive web design
- Tips for responsive CSS
  - Media Query
  - Flexbox and Grid
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- Making easy responsiveness layouts

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Responsive Web Design

- When you use CSS and HTML to resize, hide, shrink, enlarge, or move the content to make it look good on any screen.
- A CSS that offers better user experience across different devices and screen sizes.



Ref: https://www.seguetech.com/css3-supports-responsive-design/

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Approaches to Responsive Web Design

- Create an miniaturized version of the larger desktop view first. The miniaturized layout may not provide full experience as some links, text, graphics, and other elements might be too small to adequately interact with.
- Create multiple layouts – a fixed width for large and medium screens and fluid widths for smaller screens.

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Responsive CSS Tips

- Use media Query.
- Use Relative Units.
- Use Flexbox and Grid.
- Minimize the use of absolute positioning.
- Optimize images: compressing them and reducing their file size.
- Test your sytles.

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Media Queries

- CSS2 introduced media types to enable different style rules to be specified for different devices.
- CSS3 introduced media queries to expand on the concept of media types in CSS2, they create more precise rules.
- Media queries allow you to define rules that only apply to certain conditions, such as the screen size, orientation, resolution or color depth of the device.
- You can specify the minimum or maximum width of the screen and apply styles accordingly.

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Media Queries

- This example applies a different style to a web page when it is viewed on a mobile device with a small screen size. It sets the font size to 14px when the screen size is less than 768 pixels.

```css
@media only screen and (max-width: 768px) {
    body {
        font-size: 14px;
    }
}
```

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Media Queries

- In HTML

  <head>:
  <link rel="stylesheet" media =
  "screen and (max-width:640px)"
  Href="mobile_device.css"/>

- In CSS

```scss
@media only screen and (max-width: 768px) {
    /* Styles for screens less than 768px */
}

@media only screen and (min-width: 769px) {
    /* Styles for screens greater than or equal to 769px */
}
```

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Media Queries

```
@media screen and (max-width:640px) {
... style rules
}
@media all {
body {background-color : blue;}
}
@media print {
body {background-color : white;}
.donotdisplay {display: none;}
}
```

@media Screen – for computer screen, tablets, mobile
@media all – for all media type devices
@media Print – for print preview mode

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Use Flexbox and Grid

- They are used for customization.
- They let you specify how elements should be positioned and sized, making it easier to create layouts that adapt to different screen sizes.

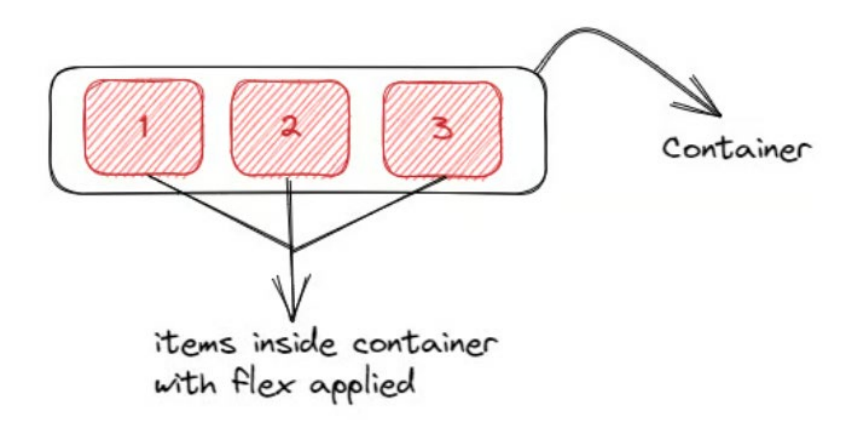


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Flexbox

```
<div class="container">
  <div id="one">1</div>
  <div id="two">2</div>
  <div id="three">3</div>
</div>

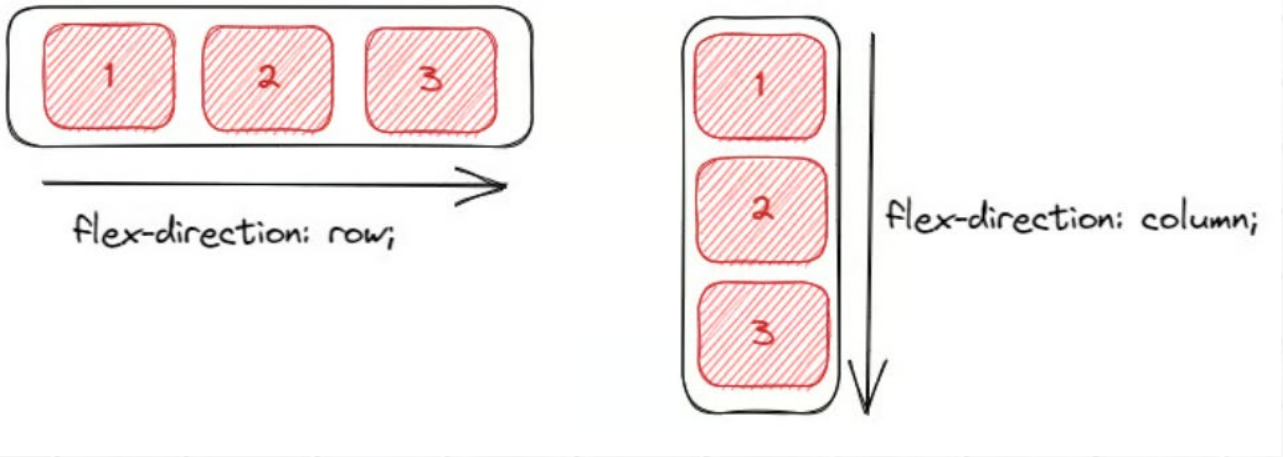
<style>
  body {
    background-color: black;
  }
  .container {
    display: flex;
  }
  #one, #two, #three {
    color: white;
    background-color: #333;
    padding: 10px;
    margin: 10px;
    border-radius: 5px;
  }
</style>
```



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More on Flexbox

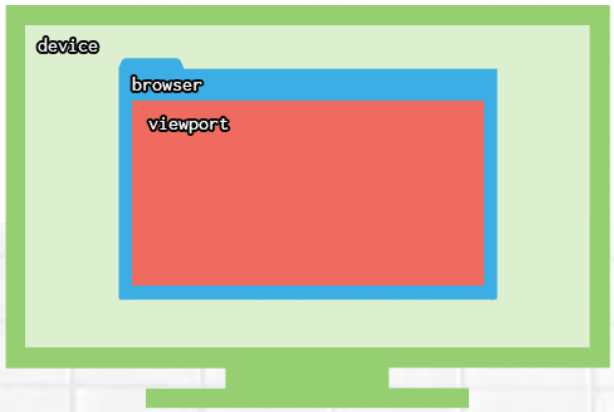
```
<style>
  .container {
    display: flex;
    flex-direction: column; /* or row */
  }
</style>
```



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ViewPort

- Viewport defines the area of a webpage which is visible to users.
- Its dimensions controls how elements will appear on the webpage.
- Use viewport meta tag in the head action of your HTML document, to define the size and scale of the viewport for optimal display on various devices.



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ViewPort - Example

```html
<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <style>
      body {
        background-color: black;
      }
    </style>
  </head>
  <body>
    <!-- Your content here -->
  </body>
</html>
```

The viewport meta data instructs the page to automatically match the width of the device

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Wake brain

- Which of the following best describes a mobile-first approach to responsive web design?



Designing for desktop screens first and the scaling down for smaller screens.



Designing for mobile screens first then scaling up for larger screens.



Designing separate websites for mobile and desktop screens.



Designing websites without considering screen size and device type.

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Wake brain

- Which of the following is NOT a common approach to creating responsive web design layouts?


Flexible grid systems


CSS media queries


Fixed-width layouts


Fluid layouts

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Wake brain

- Which of the following code snippets is an example of a media query that applies styles only when the viewport width is 600 pixels or more?



@media screen and (max-width:600px) {..}



@media screen and (min-width:600px) {..}



@media screen and (width:600px) {..}



@media screen and (device-width:600px) {..}

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Thank You

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